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CSNB544/CSNB5123, Semester 2, 2024/2025



COLLEGE OF COMPUTING AND INFORMATICS PUTRAJAYA CAMPUS

FINAL PROJECT

SEMESTER 2 2024/2025

PROGRAMME : Bachelor of Computer Science (Software Engineering) (Hons.), Bachelor of Computer Science (Systems and Networking) (Hons.), Bachelor of Computer Science (Cyber Security) (Hons.), Bachelor of Information Technology (Information Systems) (Hons.), Bachelor of Information Technology (Graphics and Multimedia) (Hons.)

SUBJECT CODE : CSNB544/CSNB5123

SUBJECT : Mobile Application Development

DATE : March 2025 (Week 4)

DURATION : 14 Weeks

INSTRUCTIONS TO CANDIDATES:

1. The total marks for this final project assessment is 100.
2. There are **TWO (2) SECTIONS** to this assessment: Section A and Section B.
3. This is an individual assessment.

***THIS QUESTION PAPER CONSISTS OF TEN (10) PRINTED PAGES
INCLUDING THIS COVER PAGE***

Instructions:

The theme for this semester is **Physical Health / Mental Health**

Develop a mobile application according to the theme assigned to you. The mobile application should contain the following components:

- A related icon to represent your application.
- A splash screen to be displayed when the application is launched.
- An about page to describe the application and developer.
- Some pages to present your application features.
- Include storage: File OR Database

You are required to use **Visual Studio Tools for .NET MAUI (Windows)** or **Visual Studio 2022 and above for Mac (macOS)** as the development tool. The application to be developed should be written in a **.NET MAUI framework** and can be deployed on Android, iOS, or both. For the database, if you are using a local database, you need to use **SQLite**. If you are using an online database, you need to use **Google Firebase**.

Your application must be original, not inherited or replicated from any existing application, not inherited from any Final Year Project, and it must be different from each other.

You should submit the following materials before the presentation session:

- **ONE (1)** PDF file - The whole question paper with completed answers in Section A
- **ONE (1)** .zip file that contains the answer for Section A (.pdf file), the Project folder URL containing the Project Files (source codes and APK files) and a video of the project demonstration.

This project is an individual assessment. Any similarity/plagiarism will result in zero mark for the whole project.

You are **NOT ALLOWED** to reuse the BMI calculator as a part of your application since the BMI application was used as part of your lab work.

You are required to present your work according to the presentation session assigned to you. Failure to do so would cause you to be considered absent from taking the final presentation and zero mark will be given. The presentation shall be done online using Microsoft Teams and the video camera must be turned on at all times during the demonstration session. Failure to do the presentation will result in failure for the entire portion of the project.

SECTION A: APPLICATION REPORT (7 QUESTIONS, 30 MARKS)**Instruction: Please complete the report.**

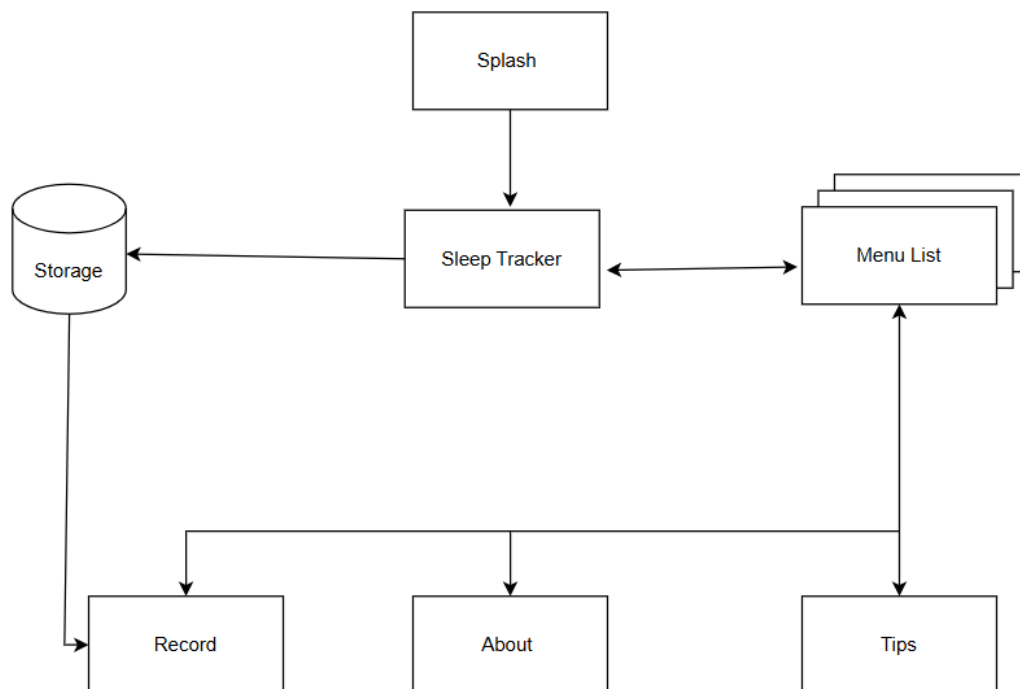
Write a report based on the developed application by completing the following table.

1. Application name	REMinder
2. Application description	REMinder is a mobile application designed to enhance users' sleep quality through effective sleep tracking, expert-backed advice, and personalized sleep logging. By helping users understand their sleep patterns and adopt better bedtime habits, REMinder supports healthier and more consistent rest.
3. Objectives	To improve sleep quality by tracking sleep patterns, providing expert tips, and logging daily sleep records.
4. Target user	People who experience inconsistent sleep habits, those that want to improve their sleep cycle, and everyone that wants to enhance their general sleep hygiene. Users can be students or working citizens with unhealthy sleep schedules.
5. List of application features with its description	1. Sleep Tracker • Users can input their bedtime and wake-up time. • The app calculates the total sleep duration. • Displays a sleep quality rating (Good, Average, Bad) based on hours slept. • Data is saved along with the date to Firebase for future tracking. 2. Sleep Log • Retrieves and displays saved sleep records from Firebase. • Each entry includes the date, total sleep hours, and sleep quality. • Organized in a table layout for clear trend analysis.

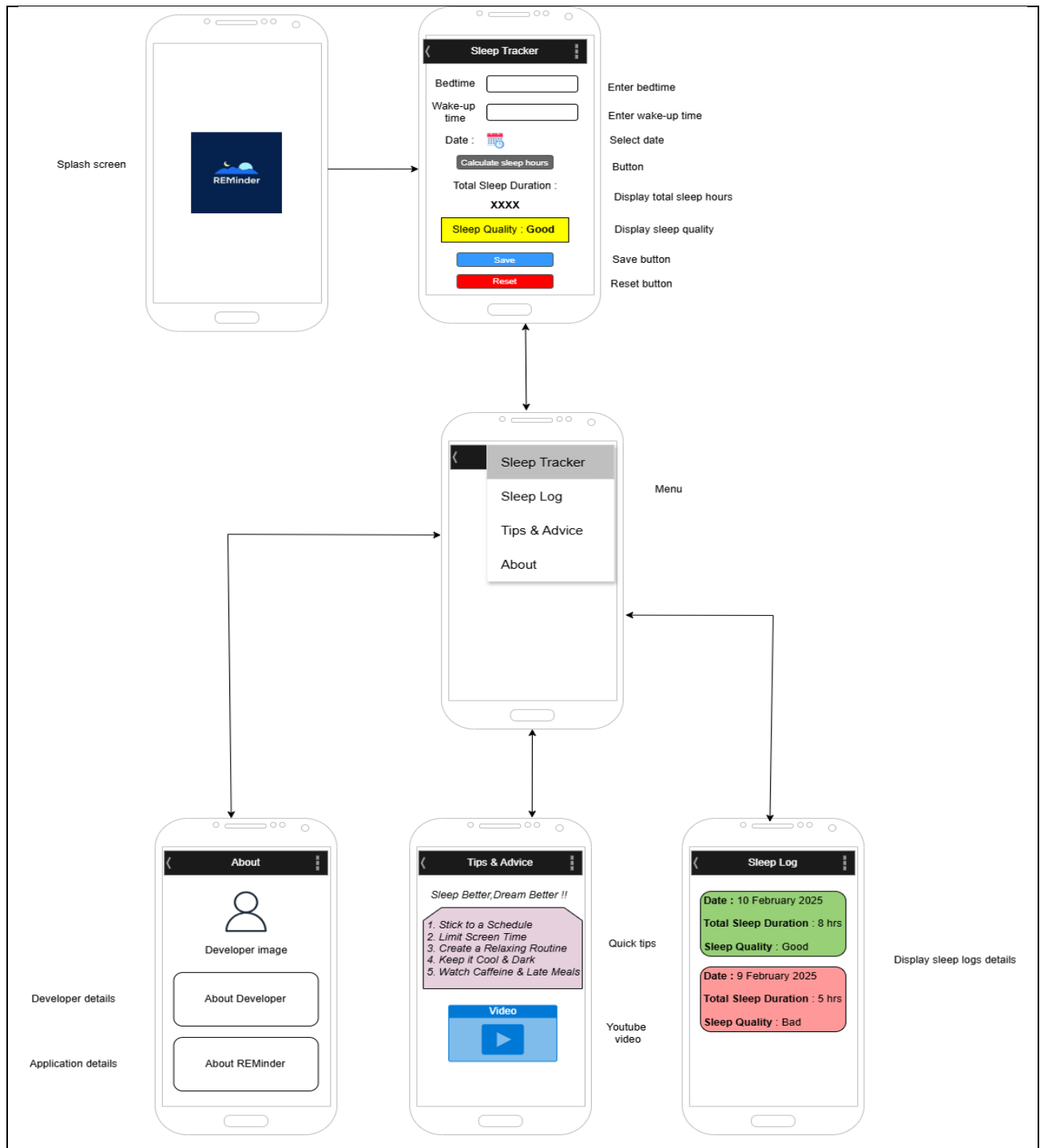
3. Tips & Advice

- Provides simple, actionable tips to enhance sleep hygiene (e.g., limit screen time, maintain consistent schedule).
- Includes embedded YouTube videos offering relaxation guidance and expert sleep advice.

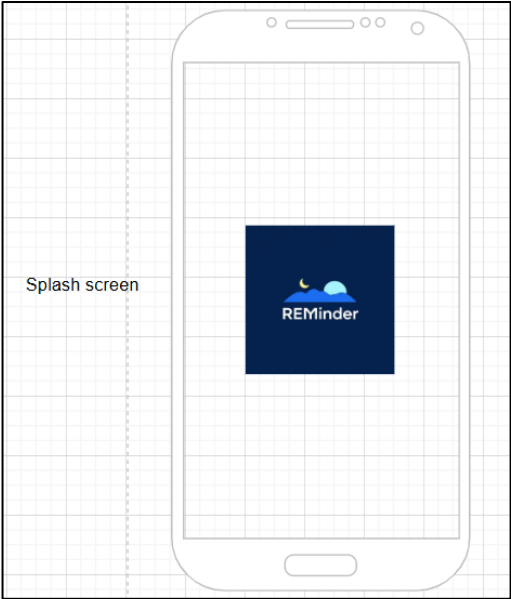
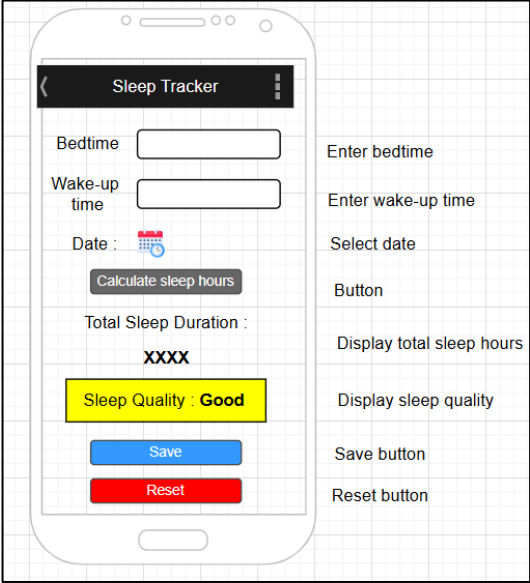
4. Draw a complete overall screens map of your application.

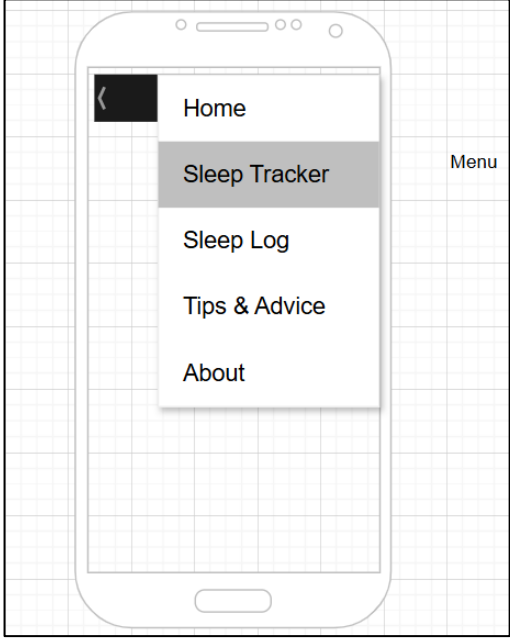
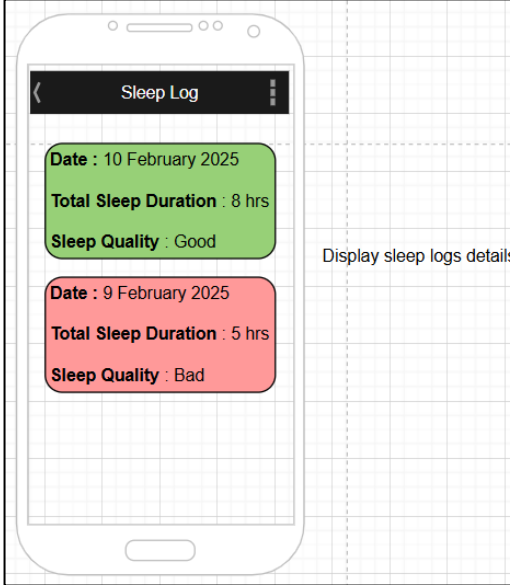


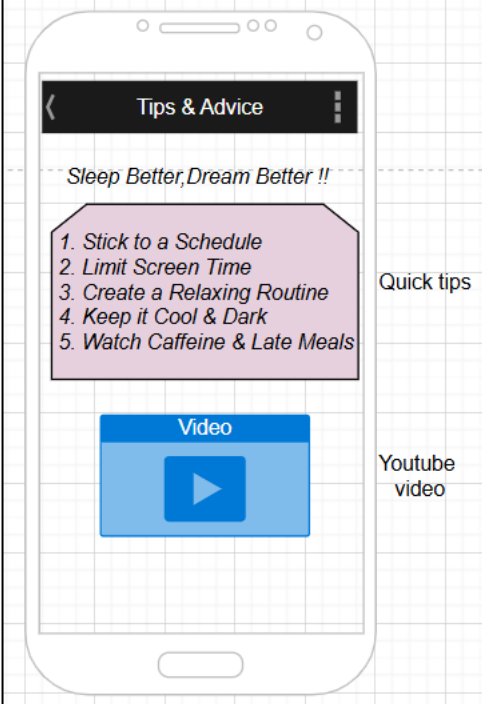
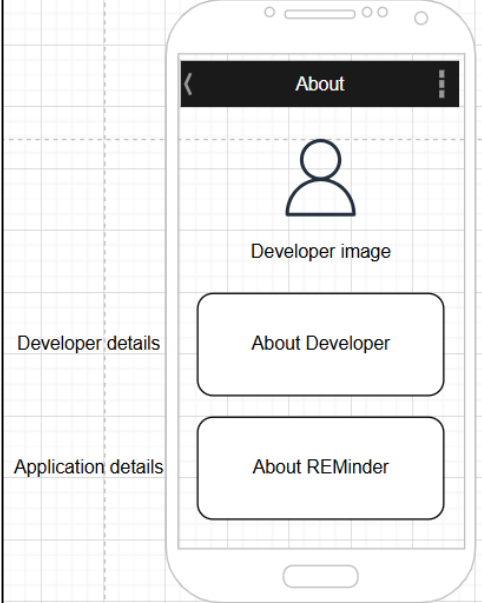
5. Sketch the UI design and wireframe based on the screens map.



[30 marks]

 Splash screen	Splash Screen • A logo named "REMinder" will be displayed to the user upon launching the application. • The screen features a dark blue background with a moon and cloud icon, symbolizing sleep and relaxation.
 Sleep Tracker Bedtime Enter bedtime Wake-up time Enter wake-up time Date : Select date Calculate sleep hours Button Total Sleep Duration : XXXX Display total sleep hours Sleep Quality : Good Display sleep quality Save Save button Reset Reset button	Sleep Tracker Page • Users can enter bedtime and wake-up time in designated text fields. • A date selector allows users to choose the specific night they are tracking. • A "Calculate Sleep Hours" button computes the total sleep duration based on input. • The total sleep duration is displayed after the calculation. • A sleep quality indicator (Good, Average, or Bad) is shown based on sleep hours. • Users can save their sleep record using the "Save" button. • A "Reset" button allows users to clear all entered data.

 A wireframe of a mobile app menu screen. It features a smartphone outline with a grid background. A dark header bar at the top contains a back arrow and the word 'Home'. Below this, a list of menu items is displayed: 'Home', 'Sleep Tracker' (highlighted with a grey background), 'Sleep Log', 'Tips & Advice', and 'About'. A label 'Menu' is placed to the right of the list.	Menu List • Provides easy navigation between different sections of the app. • Includes options for: • Home • Sleep Tracker • Sleep Log • Tips & Advice • About • The currently selected menu item (Sleep Tracker in this case) is highlighted.
 A wireframe of a mobile app 'Sleep Log' screen. It features a smartphone outline with a grid background. A dark header bar at the top contains a back arrow and the text 'Sleep Log'. Below the header, two log entries are displayed. The first entry is green and shows 'Date : 10 February 2025', 'Total Sleep Duration : 8 hrs', and 'Sleep Quality : Good'. The second entry is red and shows 'Date : 9 February 2025', 'Total Sleep Duration : 5 hrs', and 'Sleep Quality : Bad'. A label 'Display sleep logs details' is placed to the right of the logs.	Sleep Log Page • Displays previously tracked sleep records in a visually organized format. • Each log contains: ○ Date of sleep record ○ Total sleep duration in hours ○ Sleep quality indicator (Good, Bad), color-coded for clarity. • Helps users track trends in their sleep patterns over time.

	Tips & Advice Page - Displays quick sleep improvement tips to help users enhance their sleep quality. - The tips include: - Stick to a Schedule - Limit Screen Time - Create a Relaxing Routine - Keep It Cool & Dark - Watch Caffeine & Late Meals - Users can also watch an embedded YouTube video for additional sleep-related advice.
	About Page - Provides information about the app and its developer. - Displays a developer profile image. - Users can tap on: "About Developer" to learn more about the creator of the app. "About REMinder" to view details about the app's purpose, version, and functionality.

SECTION B: APPLICATION DEMONSTRATION (1 QUESTION, 70 MARKS)

Instruction: Please demonstrate the developed application.

Develop a mobile application based on the proposed idea in Milestone 1 (M1) and design idea in Milestone 2 (M2). Demonstrate your application to the lecturer and panel during the session assigned to you. You are required to have the whole project solution during the demonstration session. The demonstration **SHOULD NOT EXCEED** 30 minutes including the question-and-answer session. Failure to demonstrate your mobile application during the allocated time session will cause you to get zero for the demonstration marks and will affect the overall marks for the project evaluation.

[70 marks]

Table 1: Rubric for Report Assessment

SECTION A: REPORT					
Criteria	0	0.5	1	N/A	Marks
1. Application Name	None	Unsuitable name chosen for the app	Suitable name chosen for the app		
Criteria	0	1-2	3-4	5	Marks
2. Application description	None	Minimal or inaccurate statements to describe the application	Only brief description of the application	Detail description of the application	
3. Application Objectives	None	Minimal or inaccurate or unachievable objectives stated for the application	Sufficient objectives but inaccurately stated or not fully achievable	Sufficient objectives, accurately stated and achievable	
Criteria	0	0.5	1	N/A	Marks
4. Application target users	None	Stated the target users but without sufficient description	Well describe the target users		
Criteria	0	1-2	3-4	5	Marks
5. Application features	None	Minimal or not fully supporting objectives or irrelevant features for the application	Sufficient features but some are irrelevant or supporting objectives	Sufficient features, relevant, and supporting objectives	
6. Application screens map	None	Minimal or not fully supporting objectives or irrelevant activities and flow for the application with or without storage.	Sufficient activities and flows but some are irrelevant/ supporting objectives with or without storage.	Sufficient activities, relevant flows, and supporting objectives with storage included.	
Criteria	0	1-3	4-7	8	Marks
7. Application Sketch and Wireframe	None	Minimal sketch or irrelevant elements for the application. Poor navigation or not/partially tally with the screens map.	Sufficient sketch but some are irrelevant elements or supporting objectives. Acceptable navigation or partially tally with the screens map with minor error.	Sufficient activities, relevant elements, and supporting objectives with storage included. Acceptable navigation or tally with the screens map with none or very minimal error.	
			Total Section A [30 Marks]		

Table 2: Rubric for Application Demonstration Assessment (Lecturer)

SECTION B1: APPLICATION DEMONSTRATION (LECTURER)					
Criteria	0	0.5	1	2	Marks
1. Icon	None	Unsuitable icon chosen for the app	Suitable icon chosen for the app		
2. Splash screen	None		Included in the application with no logo	Included in the application with the application logo	
3. About page	None		Partially describe the application and developer	Completely describe the application and developer	
Criteria	0	1-9	10-19	20	Marks
4. Activities completion	None	Less than 50% completion of overall planned activities.	Less than 90% completion of overall planned activities.	Around 90-100% completion of overall planned activities.	
5. Content – Visual Control	None	Most information in the activity is confusing or irrelevant. Some of the activities do not support the objectives of the application.	Most information in the activity is relevant. Only parts of the activities support the objectives of the application.	All information is accurate and was delivered effectively. Objectives achieved.	
Criteria	0	1-2	3-4	5	Marks
6. Application navigation	None	Confusing navigation. Some of it follows the screens map, and wireframe.	Acceptable navigation. Follows the screens map, and wireframe.	Clear and suitable navigation. Follows the screens map, and wireframe.	
7. Application layout	None	Majority of the layout missing from the sketch. Layout designs are confusing.	Some of the layout matched the sketch. Some layout design support user affordance.	Whole layout design is tally with the sketch and support user affordance.	
8. Design Attractiveness	None	Majority activity design looks dull with full of texts. Lack of contrast between background and text colors made text hard to read.	Part of the activity design looks attractive, acceptable contrast, and informative with some suitable images added.	Whole activity design looks attractive, good color choices and informative with suitable images added.	
9. Application of .NET MAUI APIs	None	Applies minimal gestures or .NET MAUI API	Applies sufficient gestures or .NET MAUI API	Applies both gestures and .NET MAUI API	

10. Storage and data binding	None	Poor storage type and management or no data binding.	Suitable storage type chosen but not well organized. Applies data binding.	Suitable storage type chosen and logically organized. Applies data binding.	
			Total Section B1 (Lecturer) [70 Marks]		
			Average Section B1 (Lecturer) Total Section B1 (Lecturer)/2 [35 Marks]		

Table 3: Rubric for Application Demonstration Assessment (Panel)

SECTION B2: APPLICATION DEMONSTRATION (PANEL)					
Criteria	0	0.5	1	2	Marks
1. Icon	None	Unsuitable icon chosen for the app	Suitable icon chosen for the app		
2. Splash screen	None		Included in the application with no logo	Included in the application with the application logo	
3. About page	None		Partially describe the application and developer	Completely describe the application and developer	
Criteria	0	1-9	10-19	20	Marks
4. Activities completion	None	Less than 50% completion of overall planned activities.	Less than 90% completion of overall planned activities.	Around 90-100% completion of overall planned activities.	
5. Content – Visual Control	None	Most information in the activity is confusing or irrelevant. Some of the activities do not support the objectives of the application.	Most information in the activity is relevant. Only parts of the activities support the objectives of the application.	All information is accurate and was delivered effectively. Objectives achieved.	
Criteria	0	1-2	3-4	5	Marks
6. Application navigation	None	Confusing navigation. Some of it follows the screens map, and wireframe.	Acceptable navigation. Follows the screens map, and wireframe.	Clear and suitable navigation. Follows the screens map, and wireframe.	
7. Application layout	None	Majority of the layout missing from the sketch. Layout designs are confusing.	Some of the layout matched the sketch. Some layout design support user affordance.	Whole layout design is tally with the sketch and support user affordance.	
8. Design Attractiveness	None	Majority activity design looks dull with full of texts. Lack of contrast between background and text colors made text hard to read.	Part of the activity design looks attractive, acceptable contrast, and informative with some suitable images added.	Whole activity design looks attractive, good color choices and informative with suitable images added.	
9. Application of .NET MAUI APIs	None	Applies minimal gestures or .NET MAUI API	Applies sufficient gestures or .NET MAUI API	Applies both gestures and .NET MAUI API	

10. Storage and data binding	None	Poor storage type and management or no data binding.	Suitable storage type chosen but not well organized. Applies data binding.	Suitable storage type chosen and logically organized. Applies data binding.	
			Total Section B2 (Panel) [70 Marks]		
			Average Section B2 (Panel) Total Section B2 (Panel)/2 [35 Marks]		

Table 4: Marks Allocation Summary (Section A + Section B1 + Section B2)

Marks Allocation Summary	
	Marks
Section A: Report (Lecturer) [30 Marks]	
Section B1: Application Demonstration (Lecturer) [35 Marks]	
Section B2: Application Demonstration (Panel) [35 Marks]	
Total Marks [100 Marks]	

---End of Questions---